2025年5月10日大陆雅思纸笔考试听力（机经原题）

Part 4

You will hear a part of a lecture about Soundscapes. First, you have some time to look at questions 31 to 40.

Now listen carefully and answer questions 31 to 40.

Today, we'll delve into the interdisciplinary field of soundscape ecology, a discipline that deciphers the complex relationships between sound and ecosystem health. The concept of a soundscape—defined as the acoustic environment of a given area—was first introduced by Murray Schafer, a famous Canadian (**31) composer** . Schafer's work in the 1970s revolutionised how we perceive urban and natural environments. He argued that soundscapes are not random noise but layered compositions with inherent (**32) meaning** that could be explained in cultural values and ecological interactions. For instance, the dawn chorus of birds in a rainforest isn't just pleasant background sound; it encodes information about species diversity and territorial dynamics.

To systematically study soundscapes, ecologists divide them into three fundamental categories based on their sources. The first category, **biophony**, encompasses sounds produced by living organisms, or creatures. In Australian ecosystems, for example, this includes the croaking of (**33) frogs** in Kakadu's wetlands during monsoon season. They use distinct frequency bands to avoid vocal overlap, ensuring their mating calls are heard.

The second category, **geophony**, refers to non-living natural sounds. A striking example is the resonance of (**34) rain** on the broad leaves of Daintree Rainforest flora—a phenomenon so rhythmic that Indigenous communities historically used it to predict seasonal changes.

Finally, **anthrophony** captures human-generated sounds, from Sydney's tram networks to agricultural machinery in the Murray-Darling Basin. Unlike biophony and geophony, anthrophony often disrupts natural acoustic patterns, masking critical animal communication.

Dr. Bernie Krause, whose background in music profoundly influenced his ecological research, uncovered several groundbreaking principles in soundscape analysis. Through decades of fieldwork, Krause demonstrated that every organism contributes a distinctive （**35**） **signature** to each sound in nature. For example, the harmonics of a lyrebird's mimicry in Tasmania differ measurably from those in New South Wales, allowing researchers to track population health.

Furthermore, Krause's recordings revealed that healthy ecosystems maintain balance in both (**36) loudness** and frequency. In the Kimberley region, intact ecosystems exhibit "acoustic niches"—species avoid overlapping frequencies, much like instruments in an orchestra. Conversely, degraded habitats show chaotic frequency collisions, akin to static interference.

Beyond their compositional complexity, soundscapes offer early warnings of ecological changes. Long-term acoustic monitoring in the Australian Alps detected altered bird migration timings, signalling a shift in the**(37) climate of the environment.**

Similarly, sudden drops in insect biophony, such as the silence of bees in Western Australian orchards, can warn of various invisible (**38) pollution**, like pesticides. Studies linked these "acoustic voids" to neonicotinoid pesticides, which impair insect navigation long before chemical traces appear in soil tests.

Furthermore, modern research relies heavily on advanced tools like **acoustic recorders**, which capture vast datasets across diverse habitats. These devices enable scientists to make detailed (**39) maps** in three dimensions—time, frequency, and amplitude. For instance, those of the Great Barrier Reef revealed that fish choruses peak during full moons, synchronising with spawning cycles. They also expose "sound deserts" in over-logged Tasmanian forests, where the absence of bird calls mirrors biodiversity loss.

A question often posed in public forums is: *Which organism dominates nature's soundscape?* While sperm whales and howler monkeys are contenders, the champion is a marine invertebrate—the snapping (**40) shrimp**. Colonies inhabiting Australia's Ningaloo Reef snap their claws at 218 decibels, louder than a gunshot. This cavitation bubble implosion stuns prey and creates microhabitats for other species, proving that even the tiniest organisms can shape entire ecosystems.

That is the end of Part 4. You now have half a minute to check your answers.

2025年4月12日大陆雅思纸笔考试听力(机经原题)

Part 4

You will hear a part of a lecture about research on Dolphins in the Northern Territory. First, you have some time to look at questions 31 to 40.

Now listen carefully and answer questions 31 to 40.

Good afternoon, everyone. Today, I'll be sharing insights into our ongoing dolphin research project in Australia's Northern Territory. This region is home to two fascinating yet understudied species: the Australian Snubfin dolphin and the Indo-Pacific Humpback dolphin. Despite their ecological importance, there's a surprising lack of knowledge about their behavior, habitats, and conservation needs. Let's dive into why we chose this location, the challenges we face, and what we hope to achieve.

The Northern Territory is, in many ways, an ideal location for studying dolphins. Its coastline remains largely undeveloped, which means the water quality is **clean**, and marine life is abundant.

I've spent most of my fieldwork time in the region's major river systems. These waterways are among the few in the world where **hunting** is completely banned in these rivers, which means the dolphins and other wildlife are legally protected. This makes it yet another excellent reason to conduct research in Northern Territory.

However, there are some drawbacks. One major challenge is the extreme **weather**, which complicates research efforts. Additionally, the vast estuary zones where these dolphins thrive are highly remote.

Now, let me outline the specific objectives of this study.

First, we need to identify and classify two dolphin species，the snubfin dolphin and the humpback dolphin. So far, very few genetic samples have been collected from these species, so we also want to determine whether they are **related**.

Next, we aim to map their habitats. Dolphins are highly sensitive to their surroundings, so understanding their preferred **environment**，such as water depth, salinity, and food availability，is crucial for conservation.

Another key goal is to estimate their **population** sizes. Are these dolphins thriving, or are their numbers declining?

We also want to determine whether their movements follow any patterns，for example, if they migrate seasonally.

Finally, we need to investigate how **human** activities in their habitats affects them.

As you can see, there's still much research to be done if we want to protect these dolphins. Now, let me explain the techniques we're using.

We spend long periods camping on boats, tracking dolphins to map their hotspots. This data will be crucial for future conservation and management plans for both species.

Photography plays a huge role too. By photographing the **colours** and unique markings on their dorsal fins, we can recognize individual dolphins over time. Think of it like a fingerprint，no two dorsal fins are exactly alike! This method lets us estimate population sizes and track how often specific dolphins reappear in certain areas.

We also collect skin samples using a harmless biopsy technique，which help determine a dolphin's age. Further analysis can reveal their exposure to **pollution**.

Though our project is still in its early stages, some patterns are emerging. Both Snubfin and Humpback dolphins live in small, tight-knit groups.

They have long lifespans and reproduce slowly, breeding only once every four years. This makes them vulnerable to threats, as their populations can't rebound quickly. Interestingly, their habitats overlap with areas of increasing human activity.

However, there's still so much more to uncover about these dolphins. For instance, I'm particularly interested in further investigating whether **noise** exposure might have negative impacts on dolphin populations，questions like, could it disrupt their communication or even contribute to strandings? These questions are urgent, as coastal development in the Northern Territory is accelerating.

Our work highlights the delicate balance between conservation and human progress. By understanding these dolphins' needs, we can advocate for smarter policies. There's still so much to learn, but every piece of data brings us closer to ensuring these remarkable creatures survive for generations to come. Thank you for listening, and I'm happy to take any questions!

That is the end of Part 4. You now have half a minute to check your answers.

2025年4月5日大陆雅思纸笔考试听力（机经原题）

Part 4

You will hear a part of a lecture about the history of Ancient Egyptian Medicine. First, you have some time to look at questions 31 to 40.

Now listen carefully and answer questions 31 to 40.

Good afternoon, ladies and gentlemen. Today we're going to journey back in time to explore the remarkable medical practices of ancient Egypt ——a civilization where science, magic, and daily life intertwined in fascinating ways. Let's begin with something fundamental: how they recorded their medical knowledge.

Now, if you're imagining thick textbooks like we have today, think again. The Egyptians had a very different approach. First, the harsh reality: only **12** complete medical texts survive today.

One major challenge in studying Egyptian medicine is how information was recorded. Unlike modern medical textbooks with detailed instructions, their knowledge was often fragmented. Many ingredients are simply arranged in **lists**， just names of plants or minerals without details of how to prepare or use them. This makes interpretation incredibly difficult, especially so little contextual information survives.

Doctor Jackie Campbell is a researcher in Egyptology and medical history at the University of Manchester, specializing in ancient Egyptian medicine. She found that Ancient Egyptian medicine made extensive use of aromatic spices, which served both medicinal and ritual purposes.

While common spices could be grown locally in temple gardens along the Nile, many precious therapeutic spices had to be **imported** through Egypt's vast trade networks. Records show that cinnamon was transported from Southeast Asia, and black pepper made its way from India.

She also conducts research in museum collections. Doctor Campbell regularly examines well-preserved botanical specimens in major museums worldwide, including the British Museum. And vibrant wall paintings depicting Nubian merchants unloading exotic spices at Egyptian ports. In addition, detailed temple **sculptures** showing the entire process——from spice measurement to medical preparation.

But translating these texts isn't just about naming plants. Take Dr. Campbell's recent breakthrough—he realized Egyptian physicians cared about patient experience. She has studied the ancient Egyptian medical prescriptions. And she identified numerous ingredients and additional **flavor** in that. More surprisingly，they often used unexpected ingredients like mint specifically to improve palatability. What makes Campbell's findings particularly significant is how they challenge earlier assumptions.

Let's dive into Campbell's landmark study. His team successfully identified no fewer than 248 distinct substances. These ingredients fell into several important categories，such as the animal products and the botanical materials like the willow bark. As well as the **minerals** that formed a surprisingly large portion of their medicines.

What's more, her team made a remarkable discovery while studying artifacts in the Cairo Museum's surgical collection. Among the scalpels and bone saws, they found numerous fish amulets and carvings, all deliberately placed beside healing tools.

In ancient Egyptian medical practice, dead fish, particularly Nile tilapia，were incorporated not for their medicinal properties but as **symbols**. They often represented power that carved amulets shaped like Nile perch were placed on wounds, believed to transfer the river's life force into the body.

When it comes to ancient Egypt, mummies are undoubtedly one of the most iconic and fascinating aspects that immediately come to mind. The ancient Egyptians developed an incredibly sophisticated mummification process over thousands of years.

So, another groundbreaking discovery reveals that ancient Egyptian physicians employed sugar as one of their primary preservative agents in both medical treatment and mortuary practices, demonstrating remarkably advanced understanding of organic chemistry nearly 3,500 years before modern science confirmed its antimicrobial properties.

But the ancient Egyptians did not recognize the importance to maintain their medicines clean like we do today. They cared about dirt you could see，like dust or bugs. And they followed religious rules about purity，but they didn't know about invisible germs.

Although the ancient Egyptians' limited understanding of cleanliness could have negatively impacted wound treatment, their physicians actually developed remarkably effective wound care methods through centuries of practical experience.

They skillfully utilized naturally available materials with antimicrobial properties，for instance, certain **metals** were crushed into powder and applied to open wounds to prevent infection, while lead compounds were mixed with oils to reduce inflammation.

Most notably, many doctors would carefully coat injuries with thick layers of raw **honey**, which we now know contains powerful natural antibacterial elements that kill harmful microorganisms. Archaeological evidence from medical papyri and tomb paintings shows they combined these substances with clean linen bandages, creating treatment protocols that successfully healed wounds despite their incomplete knowledge of germ theory.

That is the end of Part 4. You now have half a minute to check your answers.